

BSc (Hons) in Information Technology Specialized in AI
IT3012 : Intelligent Agents
Year 3 - Semester 1 - 2026 Semester 2

Objective & Lecture Mapping: Build and compare a Simple Reflex Agent and a Model-Based Agent, then relate their code to agent architecture and observability theory.

Part 1: Practical Implementation

- Added local wall and food sensors to the environment percept.
- Implemented SimpleReflexAgent using current IF-THEN percept rules.
- Implemented ModelBasedAgent with last action, relative position and visited-cell memory.

Part 2: Analytical Evaluation - Completed Answers

1. Why is a table-driven agent impractical for chess or a lifetime of percepts?

Answer: A table-driven agent needs an entry for every possible percept sequence. The number of sequences grows exponentially with time, so storage and lookup become infeasible in a complex environment such as chess.

2. Which lines implement condition-action rules?

Answer: In SimpleReflexAgent.sense_and_act(), the checks for food_here and wall_ahead are the conditions. The selected movement or collection responses are their corresponding actions.

3. Why can a simple reflex agent loop in a partially observable environment?

Answer: Different world states can produce the same local percept. Without percept history or an internal state, the agent repeats the same response and can cycle between cells indefinitely.

4. Identify the transition model and sensor model in the model-based agent.

Answer: _record_last_transition predicts the new relative cell from the preceding action, so it acts as the transition model. Local wall and food booleans supply the sensor model; visited_cells preserves observations and supports choosing an alternative action.

Completion and Verification

The implementation is stored in the matching Week_02 branch. Automated tests pass and the runnable visual environment remains available in visual_grid_game.py.